

STAT 20000
Chapter 3 Histograms

Example: FEV Data

The FEV data are data of a sample of 654 youths, aged 3 to 19, in the area of East Boston during middle to late 1970's. The variables include

- ▶ age: subject's age in years
- ▶ fev: lung capacity of subject, measured by **forced expiratory volume** (abbreviated as **FEV**), the amount of air an individual can exhale in the first second of forceful breath in liters
- ▶ ht: subject's height in inches
- ▶ sex: gender (0 = Female, 1 = Male)
- ▶ smoke: smoking status (0 = Nonsmoker, 1 = Smoker)

age	fev	ht	sex	smoke
9	1.708	57.0	0	0
8	1.724	67.5	0	0
7	1.720	54.5	0	0
...(omitted)...				
15	3.211	66.5	0	0

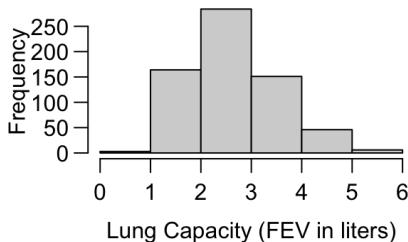
How to Make a Histogram in Frequency Scale?

How to make a histogram for fev, a measure of people's lung capacities in the fevdata?

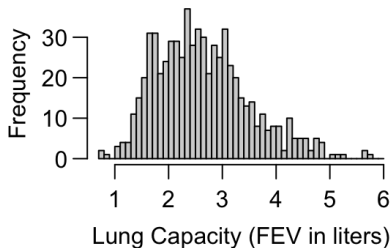
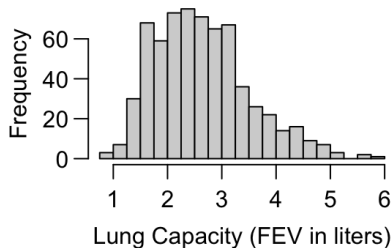
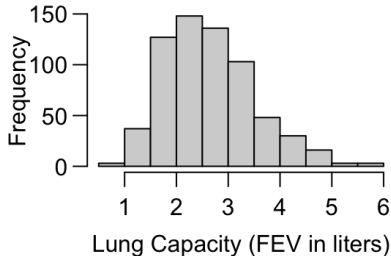
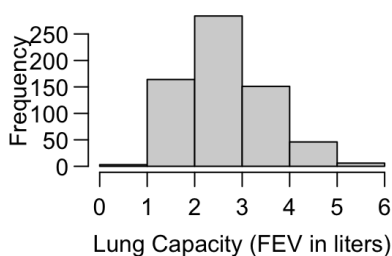
1. Divide the range of values into *class intervals* or *bins*.
2. Count the number of values in each class interval

Interval	0-1	1-2	2-3	3-4	4-5	5-6
Count	3	164	284	151	46	6

3. Draw the histogram and label the axes (with units)



Changing Binwidths Can Alter the Shape of a Histogram



Selection of Binwidth

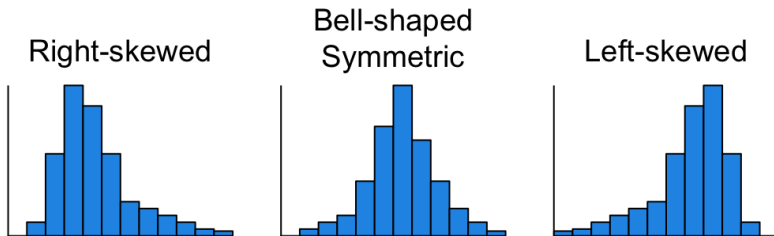
It is an iterative process — try and try again.

What binwidth should you use?

- ▶ Not too small that most bins have either 0 or 1 counts
- ▶ Not too big that you lose the details in a bin
- ▶ (There may not be a unique “perfect” bin size)

General rule: **the more observations, the more bins.**

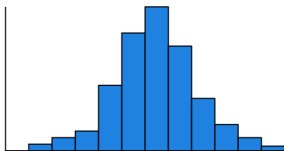
Skewness of Histograms



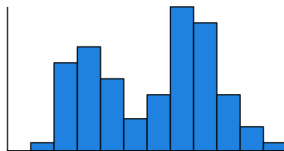
Direction of Skewness = Direction of the Longer Tail

Mode of Histograms (= Number of Peaks)

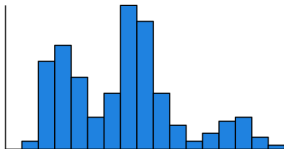
Unimodal



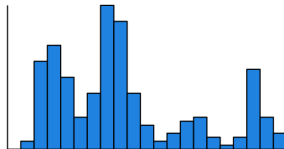
Bimodal



Trimodal



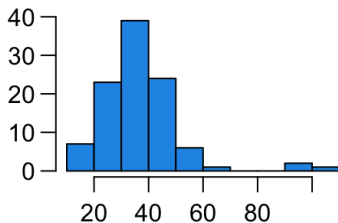
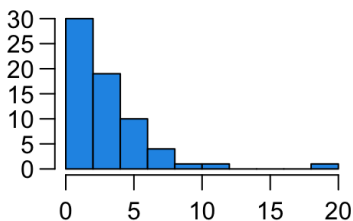
Multimodal



Two or more modes indicate data might come from two or more distinct populations.

Outliers

We can also check if there are any unusual observations or potential *outliers* from a histogram.

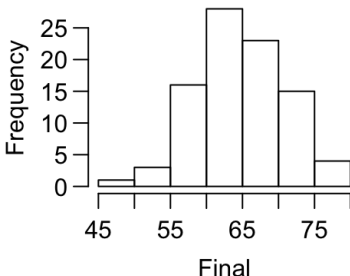
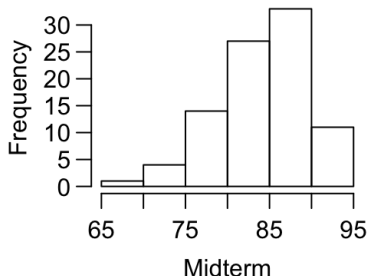


Outliers could be unusual observations or mistakes. Check them!

Comparing Histograms (1)

Here are the histograms of the midterm and the final exam scores for a class of Elementary Statistics students.

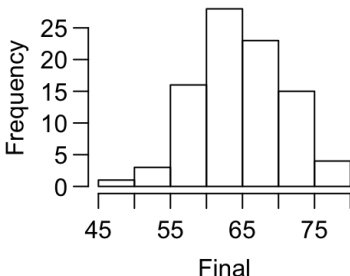
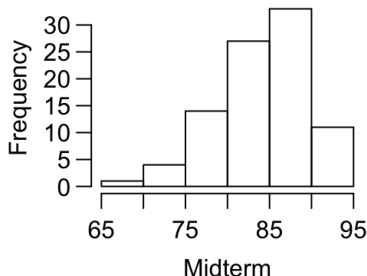
Which exam scores are generally higher, midterm or final?



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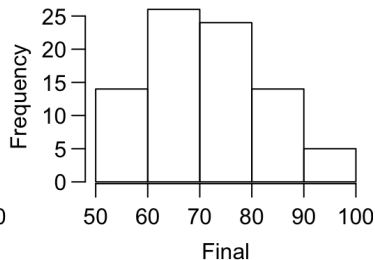
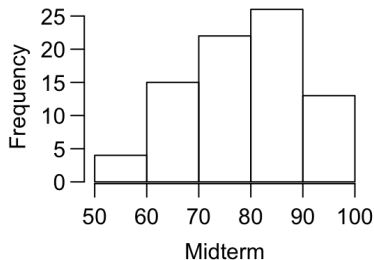
Which exam scores are generally higher, midterm or final?



Key: compare the x-axis!

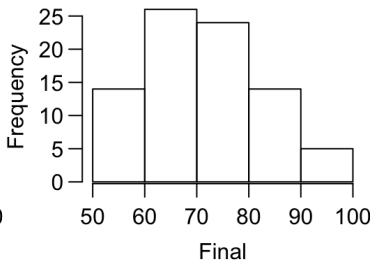
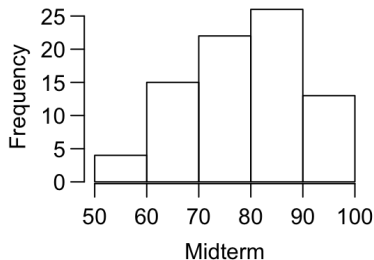
Comparing Histograms (2)

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Comparing Histograms (2)

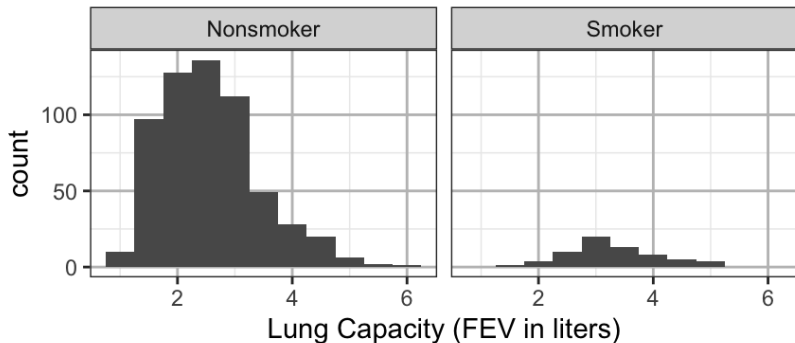
Which exam scores are generally higher, midterm or final?



Key: The average would be higher if bars for higher values have more observations, and bars for lower values have fewer observations,

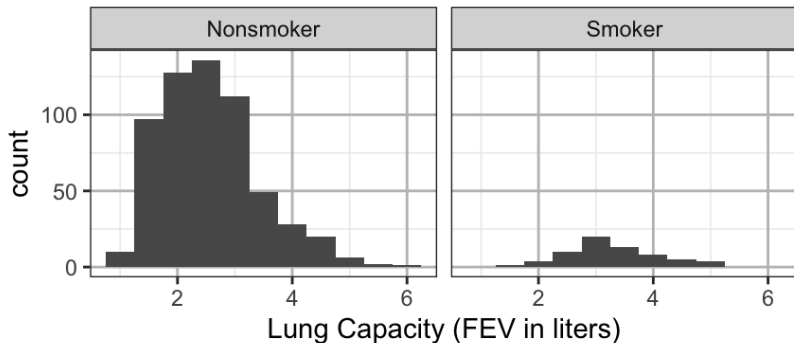
Comparing Histograms (3)

Did smokers or nonsmokers have greater lung capacities in general?



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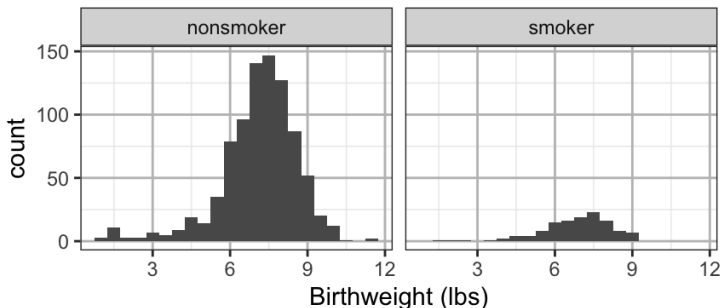


When two histograms have unequal numbers of observations, we cannot determine which one has a higher mean by comparing the heights of bars. This explains the need for histogram in density scale.

Comparing Histograms (4)

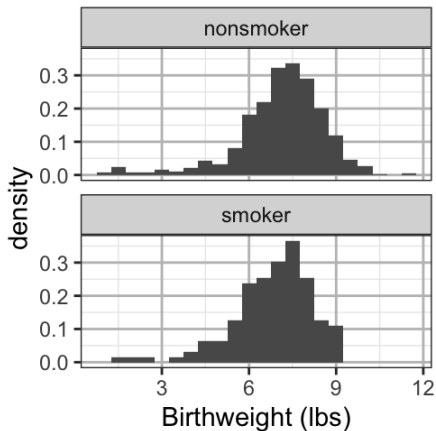
Below are the histograms of the birth weights (lbs) of 1000 randomly sampled babies born in NC in 2024, classified by mother's smoking status.

Which group had a higher birth weights in general, babies of smoking moms or those of nonsmoking moms?



Back to Smoking & Birthweight

Did babies born to smoking or nonsmoking mothers have a higher birthweights?



Histogram in Density Scale

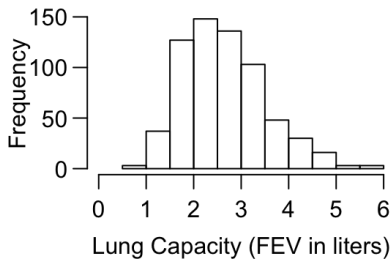
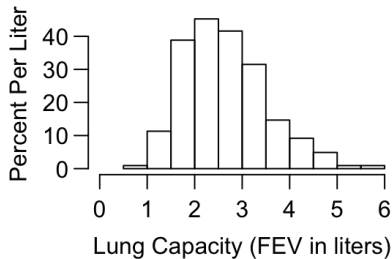
So far we only introduced histograms in *frequency* scale, of which the bar height represents the **count** of **frequency** of observations in the class interval (bin).

For histogram in *density* scale,

bar area = **percentage** of observations in the bin.

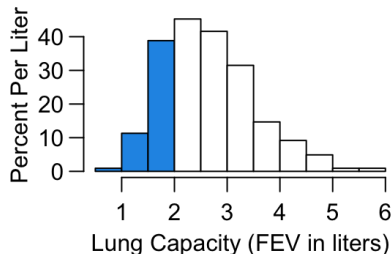
Hence bar height =
$$\frac{\% \text{ of observations in the bin}}{(\text{bin width})}$$

Interval	Count	Proportion = $\frac{\text{Count}}{\text{Total}}$	Bar Height = $\frac{\text{Proportion}}{\text{Bin Width}}$
0.5-1	3	$\frac{3}{654} \approx 0.46\%$	$\frac{0.46}{0.5} = 0.92$
1-1.5	37	$\frac{37}{654} \approx 5.66\%$	$\frac{5.66}{0.5} = 11.31$
1.5-2	127	$\frac{127}{654} \approx 19.42\%$	$\frac{19.42}{0.5} = 38.84$
2-2.5	148	$\frac{148}{654} \approx 22.63\%$	$\frac{22.63}{0.5} = 45.26$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
5.5-6	3	$\frac{3}{654} \approx 0.46\%$	$\frac{0.45}{0.5} = 0.92$
Total	654	100%	

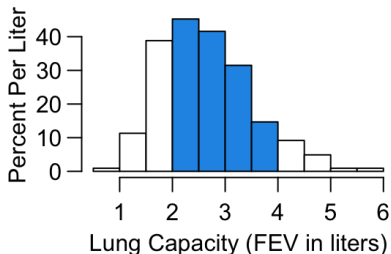


Area Under a Histogram = Proportion

In density scale,
Blue *area* = *percentage* of
subjects with FEV
between 0.5 to 2.0



Blue *area* = *percentage* of
subjects with FEV
between 2.0 and 4.0



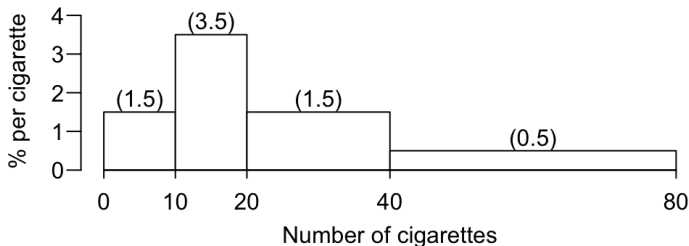
In the density scale, the total area under a histogram is 100%
(why?).

Exercise 1 (Number of Cigarettes)

The histogram below shows the number of cigarettes smoked per day for male current smokers in a public health study.

The density (bar height) is marked in parentheses.

The class intervals include the right endpoint, but not the left.



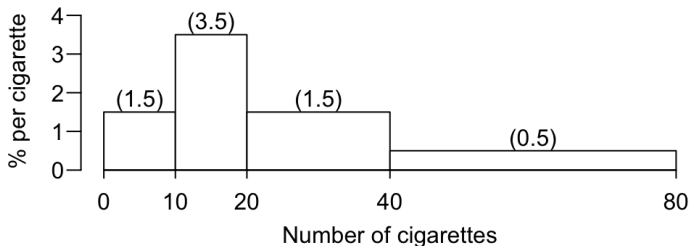
1. The percentage who smoked 10 cigarettes or less per day is around _____

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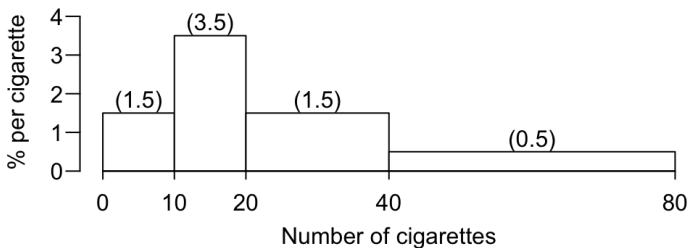
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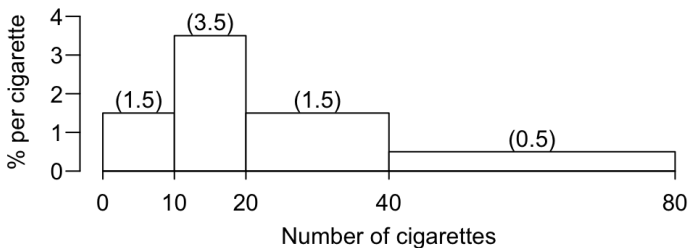
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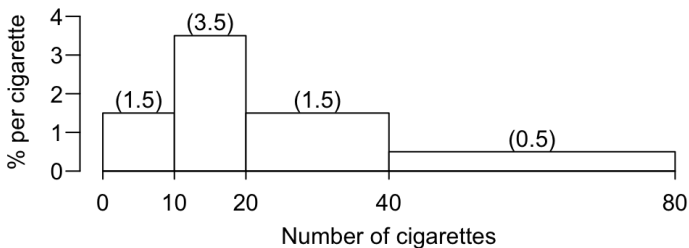
1. The percentage who smoked 10 cigarettes or less per day is around $1.5\% \times 10 = 15\%$



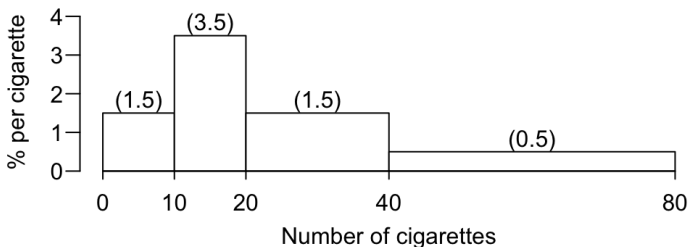
2. The percentage who smoked more than a pack a day, but not more than 2 packs, is around _____
(A pack = 20 cigarettes)
3. The percent who smoked more than a pack a day is around _____
4. The percent who smoked more than 3 packs a day is around _____



2. The percentage who smoked more than a pack a day, but not more than 2 packs, is around $1.5\% \times 20 = 30\%$
(A pack = 20 cigarettes)
3. The percent who smoked more than a pack a day is around _____
4. The percent who smoked more than 3 packs a day is around _____



2. The percentage who smoked more than a pack a day, but not more than 2 packs, is around $1.5\% \times 20 = 30\%$
(A pack = 20 cigarettes)
3. The percent who smoked more than a pack a day is around $1.5\% \times 20 + 0.5\% \times 40 = 50\%$
4. The percent who smoked more than 3 packs a day is around _____



2. The percentage who smoked more than a pack a day, but not more than 2 packs, is around $1.5\% \times 20 = 30\%$
(A pack = 20 cigarettes)
3. The percent who smoked more than a pack a day is around $1.5\% \times 20 + 0.5\% \times 40 = 50\%$
4. The percent who smoked more than 3 packs a day is around $0.5\% \times 20 = 10\%$

Exercise 2 (Wages)

Below is a histogram of weekly wages for part-time employees. Nobody earned more than \$1,000 a week. The block over the class interval \$200-\$500 is missing. How tall must it be?



- ▶ 1 unit in the horizontal scale = _____.
- ▶ Remember: the total area of a histogram is _____.
- ▶ Area of the missing block = _____.
- ▶ Height of the missing block = _____.

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- ▶ Area of the missing block = 45%.
- ▶ Height of the missing block = _____.

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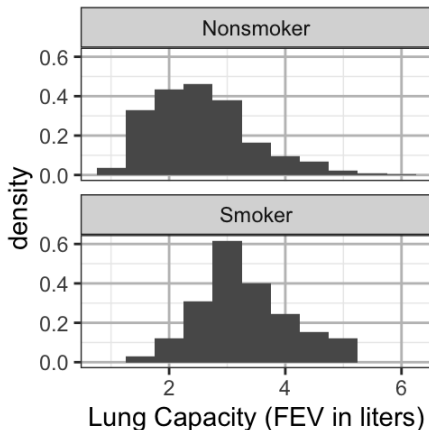


- ▶ 1 unit in the horizontal scale = \$100.
- ▶ Remember: the total area of a histogram is 100%.
- ▶ Area of the missing block = 45%.
- ▶ Height of the missing block = $45\%/3 = 15\%$.

Back to FEV Data

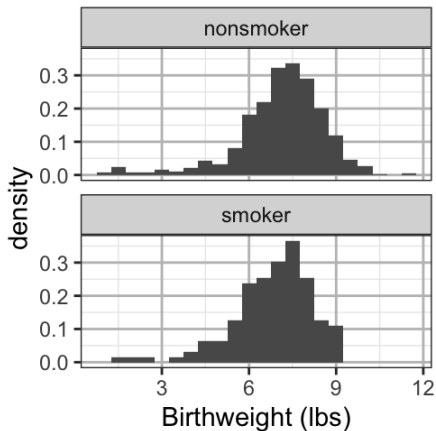
Stacking histograms vertically makes it easier to **compare the horizontal scale**.

Did smokers or nonsmokers have greater lung capacities in general?



Back to Smoking & Birthweight

Did babies born to smoking or nonsmoking mothers have a higher birthweights?



Summary of Histograms

What to Look in a Histogram?

► Shape

- symmetric or skewed (lopsided)
- number of modes (peaks)

► Outliers

► Center: Where is the “middle” of the histogram?

- typically represented by average and median

► Spread: What big the range the data spread?

- typically represented by SD and IQR (will introduce soon)

Keep in mind that

Area Under a Histogram = Percentage
